Derivative Rules - A Very Short Handout

The derivative of a function measures its instantaneous rate of change.
For a power function the rule is simple: d/dx x^n = n x^(n-1).

Examples:
  f(x) = x^2   => f'(x) = 2x
  f(x) = 3x + 1   => f'(x) = 3

The chain rule composes derivatives:
  d/dx f(g(x)) = f'(g(x)) * g'(x)

A function that is differentiable at a point is also continuous there,
but continuity does not imply differentiability. The absolute value |x|
is the standard counterexample at x = 0.

Practice: compute the derivative of x^3 + 2x at x = 1 (answer: 5).